

Program-7

Title: Deploying a Web Application on Kubernetes Using Minikube

This document describes the procedure for deploying a basic Nginx web server on a local Kubernetes cluster powered by Minikube.

1. Creating the Deployment Configuration

The deployment process begins by defining the desired application state using a Kubernetes Deployment manifest written in YAML format.

Key Configuration Elements:

kind: Deployment – Specifies that the resource being created is a Deployment.

replicas: 2 – Ensures that two identical Pods are maintained for availability and load distribution.

image: nginx:latest – Pulls the official Nginx Docker image from Docker Hub.

containerPort: 80 – Indicates that the Nginx service inside the container listens on port 80.

2. Starting and Verifying Minikube

Before deploying the application, ensure that the Minikube cluster is running and properly configured to accept Kubernetes workloads.

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
bash + v [ ] [ ] ... | [ ] [ ] x

• rv24mc081_r_mahesh_babu@fedora:~/Downloads/PRG7$ minikube start
🐳 minikube v1.37.0 on Fedora 43
🌟 Using the qemu2 driver based on existing profile
👍 Starting "minikube" primary control-plane node in "minikube" cluster
🔧 Updating the running qemu2 "minikube" VM ...
🚢 Preparing Kubernetes v1.34.0 on Docker 28.4.0 ...
🔗 Configuring bridge CNI (Container Networking Interface) ...
🔊 Verifying Kubernetes components...
  ▪ Using image gcr.io/k8s-minikube/storage-provisioner:v5
  ▪ Using image docker.io/kubernetesui/dashboard:v2.7.0
  ▪ Using image docker.io/kubernetesui/metrics-scraper:v1.0.8
💡 Some dashboard features require the metrics-server addon. To enable all features please run:

    minikube addons enable metrics-server

🌟 Enabled addons: storage-provisioner, default-storageclass, dashboard
🎉 Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
• rv24mc081_r_mahesh_babu@fedora:~/Downloads/PRG7$ minikube status
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured

• rv24mc081_r_mahesh_babu@fedora:~/Downloads/PRG7$
```

3. Deploying the Application

The deployment manifest is applied to the Kubernetes cluster using the kubectl command, which creates the Deployment and automatically spawns the required Pods.

Command:

kubectl apply -f nginx-deployment.yaml

Verification Output:

deployment.apps/nginx-deployment created

4. Checking Deployment and Pod Status

After deployment, the status of the Deployment and its associated Pods is verified to confirm that all resources are running as expected.

```
! nginx-deployment.yaml X
! nginx-deployment.yaml
1  apiVersion: apps/v1
2  kind: Deployment
3  metadata:
4    name: nginx-deployment
5  spec:
6    replicas: 2
7    selector:
8      matchLabels:
9        app: nginx
10   template:
11     metadata:
12       labels:
13         app: nginx
14     spec:
15       containers:
16       - name: nginx
17         image: nginx:latest
18         ports:
19         - containerPort: 80
20
```

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
bash + v [ ] [ ] ... [ ] [ ]

rv24mc081_r_mahesh_babu@fedora:~/Downloads/PRG7$ kubectl apply -f ./nginx-deployment.yaml
deployment.apps/nginx-deployment configured
rv24mc081_r_mahesh_babu@fedora:~/Downloads/PRG7$ kubectl get deployments
NAME                READY   UP-TO-DATE   AVAILABLE   AGE
nginx-deployment    2/2     2            2           5h21m
rv24mc081_r_mahesh_babu@fedora:~/Downloads/PRG7$ kubectl get pods
NAME                READY   STATUS    RESTARTS   AGE
nginx-deployment-6f9664446b-2599x  1/1     Running   4 (8m7s ago)  5h21m
nginx-deployment-6f9664446b-lbr2k  1/1     Running   4 (8m7s ago)  5h21m
nodejs-pod          1/1     Running   2           24m
rv24mc081_r_mahesh_babu@fedora:~/Downloads/PRG7$ kubectl expose deployment ./nginx-deployment.yaml
```

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
bash + v [ ] [ ] ... [ ] [ ]

rv24mc081_r_mahesh_babu@fedora:~/Downloads/PRG7$ kubectl get services
NAME                TYPE        CLUSTER-IP    EXTERNAL-IP  PORT(S)          AGE
kubernetes          ClusterIP   10.96.0.1     <none>       443/TCP          23h
nginx-service       NodePort    10.111.242.116 <none>       80:30465/TCP     5h23m
rv24mc081_r_mahesh_babu@fedora:~/Downloads/PRG7$ minikube service ./nginx-deployment.yaml
```

5. Accessing the Web Application

Terminal Output:

The terminal displays a URL such as **http://192.168.49.2:31858**, which can be used to access the application. Minikube attempts to automatically open this URL in the system's default web browser.

Final Browser Verification:

Visiting the provided URL successfully loads the "Welcome to nginx!" page, confirming that the Nginx web application has been deployed correctly and is accessible through the Kubernetes cluster.

